Fiscal Year:	FY 2026	**Task Last Updated:**	FY 08/27/2025
PI Name:	Weber, Richard Ph.D.		
Project Title:	Microgravity Investigation of Thermophysical Properties of Supercooled Molten Metal Oxides		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	847-612-8597
Organization Name:	Materials Development Inc.		
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City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	05/21/2017	**End Date:**	12/31/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 12/31/2025 per PI (Ed., 10/15/25) NOTE: End date changed to 05/20/2025 per NSSC information (Ed., 6/21/23) NOTE: End date changed to 05/20/2024 per NSSC information (Ed., 6/21/23) NOTE: End date changed to 05/20/2023 per NSSC information (Ed., 8/25/22) NOTE: New grant number 80NSSC18K0059 awarded with period of performance of 11/9/2017-5/20/2022; added to original grant since is continuation of the work (Ed., 5/18/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0059 ; NNX17AH73G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project is investigating the thermophysical properties of molten metal oxides. These materials are of critical importance in a wide range of applications including functional glasses, batteries, next generation propulsion systems, lasers and optical products. An increased understanding of oxide materials is of fundamental value in materials science and applied value in developing means to harness planetary materials for use in space exploration. Fundamental research provides an essential underpinning for transformational discovery and development of major new applications. This project aims to measure the density, thermal expansion coefficient, viscosity, and surface tension of a variety of molten oxides, some of which can form glass. Containerless conditions are essential in order to achieve deep supercooling without extrinsic heterogeneous nucleation of crystals. This requirement will be met by using the Japanese Aerospace Exploration Agency (JAXA) Electrostatic Levitation Furnace (ELF). The ELF instrument is an electrostatic levitator designed for work on metal oxides at temperatures up to about 2500 ºC in containerless conditions. By working in microgravity, nearly quiescent molten samples can be obtained. Induced oscillations of levitated liquid droplets is used to study their fluid properties. Image analysis reveals the oscillation amplitude and power spectrum, from which the required properties (surface tension and viscosity) are calculated. The materials of primary interest are molten metal oxides, many of which form fragile liquids that can be made into glass provided the melts can be deeply supercooled. In some cases there is already considerable data on the atomic structure of these melts. An important goal of this research is to correlate the structure and properties (in particular viscosity and density) to advance models of the processing of materials. These models are important in advancing basic science of liquids and glasses and in development of high value-added applied materials. Samples returned from the flight experiments are being investigated by a variety of techniques and in collaboration with several research groups at US universities and National Laboratories. ED. NOTE 5/18/2020: New grant number 80NSSC18K0059 awarded with period of performance of 11/9/2017-5/20/2024; added to original grant since is continuation of the work.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is investigating molten metal oxides. These materials are of critical importance in a wide range of applications including functional glasses, batteries, next generation propulsion systems, lasers and optical products. An increased understanding of oxide materials is of fundamental value in materials science and applied value in developing means to harness planetary materials for use in space exploration. Fundamental research provides an essential underpinning for transformational discovery and development of major new applications. The project work includes outreach to undergraduate students. The experiments in microgravity are measuring essential thermophysical property data that can be used to advance understanding of how to process and optimize materials performance both on Earth and for space exploration. Two patents on the work are being used to develop commercial activities from the resulting products.
	This project is investigating the thermophysical properties of molten metal oxides. These materials are of critical importance in a wide range of applications, including functional glasses, batteries, next-generation propulsion systems, lasers, and optical products. An increased understanding of oxide materials is of fundamental value in materials science, and has applied value to technologies for processing planetary materials in space exploration. Fundamental research provides an essential underpinning for transformational discovery and the development of major new applications. This project is measuring the density, thermal expansion coefficient, viscosity, and surface tension of a variety of molten oxides, some of which can form glass. Containerless conditions are essential to achieve deep supercooling without extrinsic heterogeneous nucleation of crystals. This requirement is met by using the Japanese Aerospace Exploration Agency (JAXA) Electrostatic Levitation Furnace (ELF). The ELF instrument is an electrostatic levitator designed for work on metal oxides at temperatures up to about 2500 ºC in containerless conditions. By working in microgravity, nearly quiescent molten samples can be obtained. Induced oscillations of levitated liquid droplets are used to study their fluid properties. For example, image analysis reveals the oscillation amplitude and power spectrum, from which the required properties (surface tension and viscosity) are calculated. The materials of primary interest are molten metal oxides, many of which form fragile liquids that can be made into glass, provided the melts can be deeply supercooled. In some cases, there are already considerable data on the atomic structure of these melts. An important goal of this research is to correlate the structure and properties (in particular, viscosity and density) to advance models for the processing of materials. These models advance the science of liquids and glasses and are important in the development of high value-added applied materials. Samples returned from the flight experiments are being investigated by a variety of techniques and in collaboration with several research groups at U.S. universities and national laboratories. 1. STATUS: The project team includes Materials Development Inc. (MDI) scientists and Dr. Shinji Kohara, National Institute for Materials Science, Japan and Dr. Takehiko Ishikawa, JAXA, Japan. Experiments on ELF are performed by Dr. Chihiro Koyama and coordinated by Mr. Kazuki Yukumatsu from JAXA. The grant was awarded on 5/21/2017 and assigned a new contract number in 11/2017. First-year funding was at a level sufficient to scope the work and perform preliminary ground-based experiments in preparation for a more detailed experimental campaign during the subsequent years. Second-year funding was at a level sufficient for planning and preparation for the Science Concept Review (SCR). The SCR was successfully completed in March 2020. Some minor items identified during the SCR were addressed soon afterwards. The first round of flight experiments was started in November 2021 and completed in March 2022. This work included experiments on more than 30 samples. Due to some operational issues, some samples could not be run. In an effort to complete the investigation, JAXA provided a fourth sample holder, and MDI provided additional samples. The fourth sample holder was sent to the International Space Station (ISS) on SpaceX CRS-27 (SpX-27) in 3/2023, and experiments were performed in 2/2024 and 3/2024. MDI was granted an extension of the period of performance due to delays in the implementation of ISS experiments. Starting in 8/2022, the NASA flight support staff and MDI investigators meet once a month via a Teams meeting hosted by Brandon Phillips at NASA Marshall Space Flight Center. The meeting maintains communications and provides for discussion of technical or project management matters that need to be addressed. 2. TECHNICAL ACTIVITIES:

Task Progress:

Activities during this year included: (i) running experiments on samples that were delivered in early 2025, (ii) analyzing data and recovered samples, (iii) making ground-based structure and property measurements on glass and liquid samples, and (iv) publishing results.

(i) Flight experiments were run in the JAXA ELF during the period 11/2021-3/2022, and 2/2024 and 3/2024. Initially, five samples each of nine different compositions were launched. An additional holder of samples was launched in 3/2023. These samples were run in ELF in early 2024. Six samples were launched in early 2025 and experiments in ELF were performed in August 2025.

A precision stainless steel calibration sphere (identified as SUS) was measured at the start and end of each sample holder set. The methodology for calculating sample volume compared to the calibration sphere was refined, based on a collaboration with Professor Robert Hyers and Dr. Richard Bradshaw. Following these modifications to the density analysis algorithm, all sample density analyses were rerun using mass data from recovered samples and internal porosity corrections obtained from post-flight X-ray tomography. Viscosity and surface tension data were obtained from the drop oscillation spectra. These data were reduced using programs provided by JAXA, with filter settings selected based on analysis of the error sources and effects of filter bandpass and frequency range [1]. Once the data processing approach was developed, it was applied consistently to all data sets in order to avoid introducing subjective bias.

(ii) Analyzing data is an ongoing process as new data and details are available. Much of the data has now been analyzed and in many cases published (please see list at the end of the report). [Ed. Note: See also the Cumulative Bibliography.] Tomography was performed on all samples for which density was measured. In cases where samples formed glass, corrections for any voids detected in the tomography were made.

Models are being used to help with data analysis and to help understand results. This work includes a collaboration with Professor Collin Wilkinson at Alfred University. He is an expert in modeling melt properties from first principles. Comparing modeled data with measured structure factors will provide a check on the quality of the model fit, lending additional weight to the value of the data. The model will then be used for calculating viscosity, to compare with the ELF flight measurements. Preliminary results indicate that with suitable pair potentials, the model reproduces the measured viscosity of samples that were measured in the ELF experiments.

(iii) Ground-based tests are in progress to characterize the compositions for flight and to make measurements on the samples recovered from the flight experiments. Flight samples were examined by optical microscopy to determine the overall appearance of the beads. High energy white beam X-ray tomography was performed at the Advanced Photon Source beamline 7 BM-B to determine the interior structure of the samples with a resolution of ca. 5 μm. These measurements are useful to evaluate the presence of bubbles or interior features that might affect the calculated volume (and hence, density) of glass samples. In all cases, both flight and ground-based samples were examined to enable comparison of the materials processed in microgravity and on Earth.

For prior sets of samples, non-destructive tests using x-ray and neutron scattering, x-ray tomography, and nuclear magnetic resonance (NMR) were performed. The results of measurements for molten Yb3Al5O12, one of the Superglass compositions, were also used to refine a structural model using empirical potential structure refinement (EPSR). The structural model is in good agreement with the scattering data that is now published. Analysis of the data led to the development of a new glass structure nomenclature that can be used to describe materials with an unconventional network structure. Typical glass formers have 4-coordinated cations that form tetrahedral species. Several of the more fragile liquids being investigated in this work form higher coordinated cations that develop the network structure [2]. An additional publication is in preparation on this result.

The thermodynamics of the glassy samples were investigated in the Materials of the Universe Laboratory at Arizona State University. The enthalpy of the solution was measured to obtain the heat of vitrification. The results show that the glass transition temperature and the enthalpy of vitrification of the glasses made in microgravity and on Earth are similar. The work was recently published [3].

Work is in progress at NASA Marshall Space Flight Center to add capabilities to the ground-based ESL to enable measurements of density and synthesis of glasses. This will provide additional glass samples that can be compared to those made in microgravity.

(iv) Presenting and publishing results is an important aspect of the research. Details of publications are given below. [Ed. Note: See also the Cumulative Bibiliography.]

• Publications/presentations:

Several papers have been published based on the ISS experiments. These are listed in the references section. Talks were presented at the American Society for Gravitational and Space Research (ASGSR) meeting in Puerto Rico, in December, 2024 (Weber); American Ceramic Society Glass and Optical Materials meeting in Vancouver, May 2025 (Weber, Wilke); and Physics of Non-Crystalline Solids (PNCS-17) in Tsukuba, Japan in August 2025 (Weber, Wilke). An abstract is pending for the 2025 ASGSR meeting.

• Several peer-reviewed scientific papers were published. Listing only those in the reporting period:

1. Wilke SK, Al-Rubkhi A, Koyama C, Ishikawa T, Oda H, Topper B, Tsekrekas EM, Möncke D, Alderman OL, Menon V, Rafferty J. Clark E, Kastengren AL, Benmore CJ, Ilavsky J, Neuefeind J, Kohara S, SanSoucie M, Phillips B, Weber R. "Microgravity effects on nonequilibrium melt processing of neodymium titanate: thermophysical properties, atomic structure, glass formation and crystallization." npj Microgravity. 2024 Mar 6;10(1):26. https://

2. Benmore CJ, Bogle R, Wilke SK, Weber R, Neuefeind J, Costa G. "The structure of CaO–MgO–Al2O3–SiO2 melts and glasses doped with FeOX–NiO." Journal of the American Ceramic Society. 2024 May 5;107(9):6323-33. https://

3. Wilke SK, Al-Rubkhi A, Menon V, Rafferty J, Koyama C, Ishikawa T, Oda H, Hyers RW, Bradshaw RC, Kastengren AL, Kohara S, SanSoucie M, Phillips B, Weber R. "Measuring the density, viscosity, and surface tension of molten titanates using electrostatic levitation in microgravity." Appl. Phys. Lett. 2024 Jun 24;124(26). https://

4. Wilke SK, Al-Rubkhi A, Benmore CJ, Neuefeind J, Koyama C, Ishikawa T, Shimonishi R, Weber R. "Structure of

molten ytterbium aluminum garnet." J. Chem. Phys. 2025 Mar 24;162(12):124501. https://

5. Wilke SK, Rafferty J, Al-Rubkhi A, Weber R. "Density of molten oxides measured in an aero-acoustic levitator." J. Appl. Phys. 2024 Dec 16;136(23). https://

6. J. Tolliver, S.K. Wilke, A. Alrubkhi, A. Neumann, G. Balakrishnan, R. Weber, and Brian Topper, “Nd-doped lanthanum titanate glass laser,” Opt. Components and Mats. XXII, S. Jiang & M.J.F. Digonnet, Eds. Proc. of SPIE, 13362, 1336204 (2025). https://

7. Benmore CJ, Shi C, Alderman OL, Harvey JP, Lipke D, Weber JKR. "Structure–property relations of binary ferrite melts." J. Appl. Phys. 2025 Feb 28;137(8). https://

8. Wilke SK, Benmore CJ, Youngman RE, Moulton BJ, Al-Rubkhi A, Weber R. "Effect of rare earth size on network structure and glass forming ability in binary aluminum garnets." Phys. Chem. Chem. Phys. 2025 Jun 5;27(25):13464-75. http://dx.doi.org/

9. Bonatti L, Subramani T, Wilke SK, Weber R, Navrotsky A. "Energetics of neodymium titanate glass made on Earth and in space." ACS Earth and Space Chem. 2025 May 22;9(6):1277-81. https://

Papers on the flight data are in development.

3. PLANS:

Plans for the next project period are: (i) Continue data analysis, (ii) Continue sample analysis and characterization on returned samples, (iii) Publish/present results.

We plan to seek a No Cost Extension of the project to allow analysis of samples returned from the runs in progress in August 2025.

4. REFERENCES CITED:

[1] S.K. Wilke, A. Al-Rubkhi, V. Menon, J. Rafferty, C. Koyama, T. Ishikawa, H. Oda, R.W. Hyers, R.C. Bradshaw, A.L. Kastengren, S. Kohara, M. SanSoucie, B.L. Phillips and R. Weber, “Measuring the density, viscosity, and surface tension of molten titanates using electrostatic levitation in microgravity,” Appl. Phys. Lett., 124, 264102 (2024).

[2] S.K. Wilke, C.J. Benmore, R.E. Youngman, B.J.A. Moulton, A. Al-Rubkhi and R. Weber, “Effect of rare earth size on network structure and glass forming ability in binary aluminum garnets,” Phys. Chem. Chem. Phys., 27, 13464-13475 (2025). DOI: 10.1039/d5cp01467k

[3] L. Bonatti, T. Subramani, S.K. Wilke, R. Weber, A. Navrotsky, “Energetics of neodymium titanate glass made on Earth and in space,” ACS Earth and Space Chem., 9, 1277 (2025).

Bibliography Type:	Description: (Last Updated: 06/06/2026)
Articles in Peer-reviewed Journals	Wilke SK, Rafferty J, Al-Rubkhi A, Weber R. "Density of molten oxides measured in an aero-acoustic levitator." J. Appl. Phys. 2024 Dec 16;136(23). https://doi.org/10.1063/5.0245875 , Dec-2024
Articles in Peer-reviewed Journals	Benmore CJ, Shi C, Alderman OL, Harvey JP, Lipke D, Weber JKR. "Structure–property relations of binary ferrite melts." J. Appl. Phys. 2025 Feb 28;137(8). https://doi.org/10.1063/5.0242452 , Feb-2025
Articles in Peer-reviewed Journals	Wilke SK, Benmore CJ, Youngman RE, Moulton BJ, Al-Rubkhi A, Weber R. "Effect of rare earth size on network structure and glass forming ability in binary aluminum garnets." Phys. Chem. Chem. Phys. 2025 Jun 5;27(25):13464-75. http://dx.doi.org/10.1039/D5CP01467K , Jun-2025
Articles in Peer-reviewed Journals	Bonatti L, Subramani T, Wilke SK, Weber R, Navrotsky A. "Energetics of neodymium titanate glass made on Earth and in space." ACS Earth and Space Chem. 2025 May 22;9(6):1277-81. https://doi.org/10.1021/acsearthspacechem.5c00011 , May-2025
Articles in Peer-reviewed Journals	Wilke SK, Al-Rubkhi A, Koyama C, Ishikawa T, Oda H, Topper B, Tsekrekas EM, Möncke D, Alderman OL, Menon V, Rafferty J. Clark E, Kastengren AL, Benmore CJ, Ilavsky J, Neuefeind J, Kohara S, SanSoucie M, Phillips B, Weber R. "Microgravity effects on nonequilibrium melt processing of neodymium titanate: thermophysical properties, atomic structure, glass formation and crystallization." npj Microgravity. 2024 Mar 6;10(1):26. https://doi.org/10.1038/s41526-024-00371-x , Mar-2024
Articles in Peer-reviewed Journals	Wilke SK, Al-Rubkhi A, Benmore CJ, Neuefeind J, Koyama C, Ishikawa T, Shimonishi R, Weber R. "Structure of molten ytterbium aluminum garnet." J. Chem. Phys. 2025 Mar 24;162(12):124501. https://doi.org/10.1063/5.0254095 , Mar-2025
Papers from Meeting Proceedings	Tolliver J, Wilke SK, Alrubkhi A, Neumann A, Balakrishnan G, Weber R, Topper B. "Nd-doped lanthanum titanate glass laser." SPIE OPTO, San Francisco, California, January 25-31, 2025. Proceedings. Optical Components and Materials XXII. 2025 Mar 21;13362:1336204. SPIE. https://doi.org/10.1117/12.3042163 , Mar-2025
Patents	US2025/0145313 A1 Pending. Nov-2024 Weber R, Wilke S. "Method for Glass Formation and Morphology of Products Made from Non-Equilibrium Liquids by Processing in Reduced Gravity."